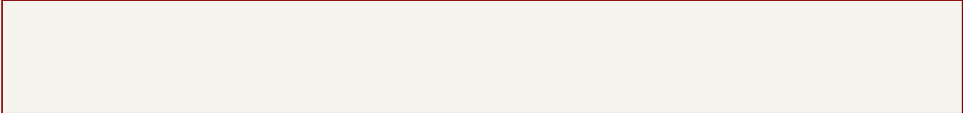


MACH3-SIMPLE-ETH - RP2350B + grblHAL, 8 axes, W5500 Ethernet

A1 with the LPT ports replaced: relays, inputs and isolated VFD port kept.

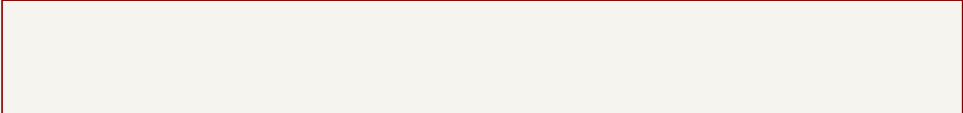
ETH-0: NOT BUILT, NOT TESTED.

Power - 24 V in, 5 V and 3.3 V made on board



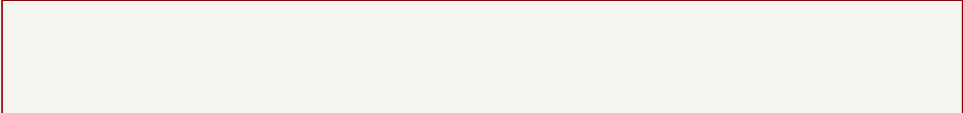
File: 01_power.kicad_sch

Ethernet - W5500 and magjack



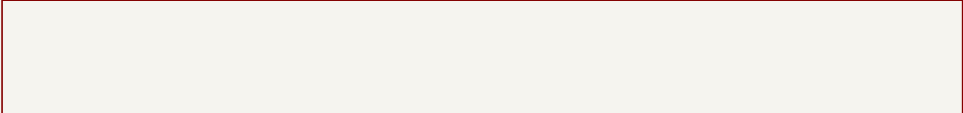
File: 03_ethernet.kicad_sch

Inputs - 15 opto-isolated 24 V channels



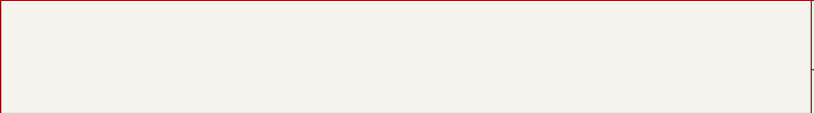
File: 05_inputs.kicad_sch

Spindle - isolated 0-10 V and FWD-REV-AUX



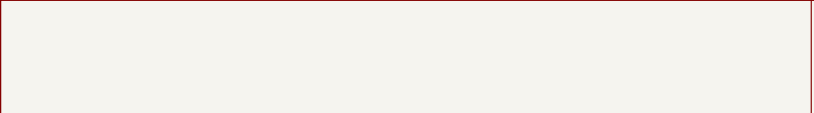
File: 07_spindle.kicad_sch

RP2350B, flash, USB, SD card



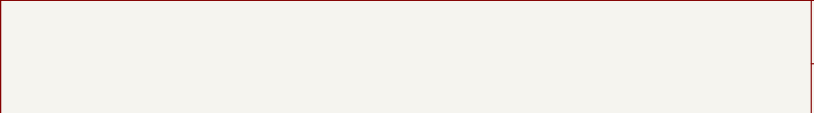
File: 02_mcu.kicad_sch

Outputs - 8 axes step/dir, enable, buffered 5 V



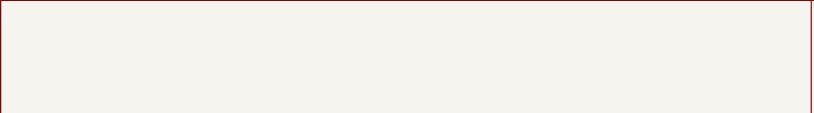
File: 04_outputs.kicad_sch

MCP23017 and six relays

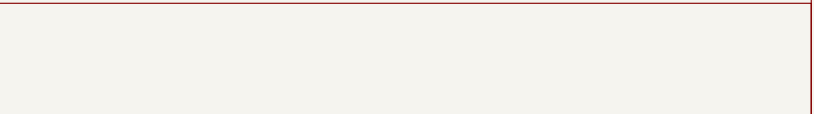


File: 06_relays.kicad_sch

RS485 VFD port and handwheel MCU



File: 08_comms.kicad_sch

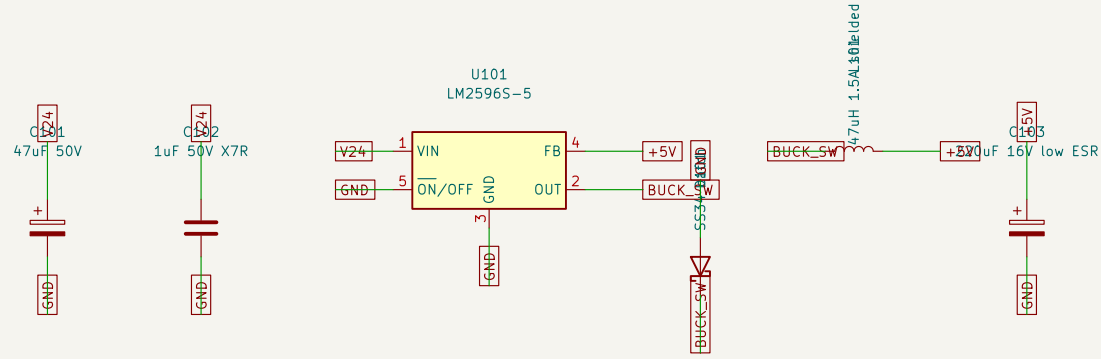


Sheet: /		
File: MACH3SIMPLEETH.kicad_sch		
Title: MACH3-SIMPLE-ETH		
Size: A4	Date:	
KiCad E.D.A. 10.0.6		Rev: ETH-0 ENGINEERING ONLY
		Id: 1/9

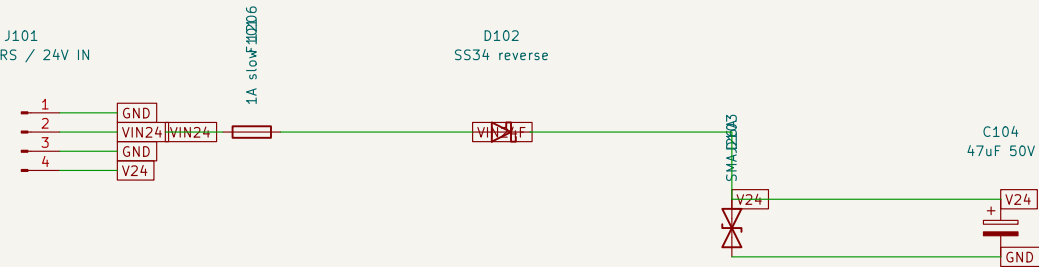
POWER. 24 V from the cabinet; 5 V and 3.3 V made on the board.

One ground, as on A1. Only the VFD port (sheet 7) is isolated.

5 V – LM2596S–5.0 BUCK, AS A1

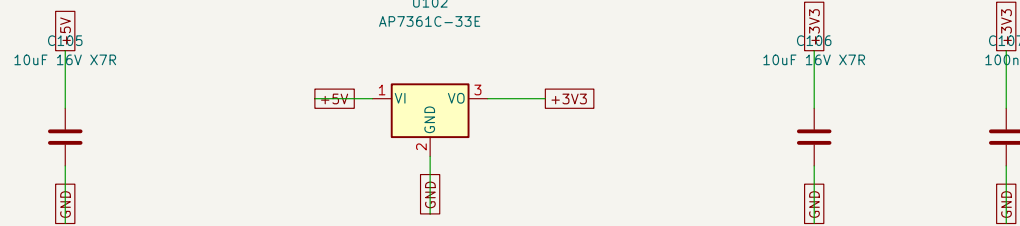


24 V IN, SENSOR SUPPLY – AS A1



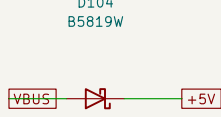
3.3 V – AP7361C LDO OFF 5 V

RP2350, W5500, SD card, MCP23017, handwheel MCU. -0.4 A worst case.

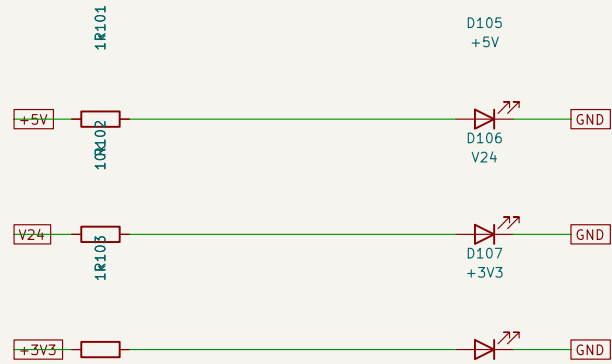


USB POWER – LOGIC ONLY, FOR FLASHING AND SETUP

VBUS feeds +5V through D104. With 24 V on, the buck holds +5V at 5.0 V and D104 is reverse biased. Relays and sensors need 24 V regardless.

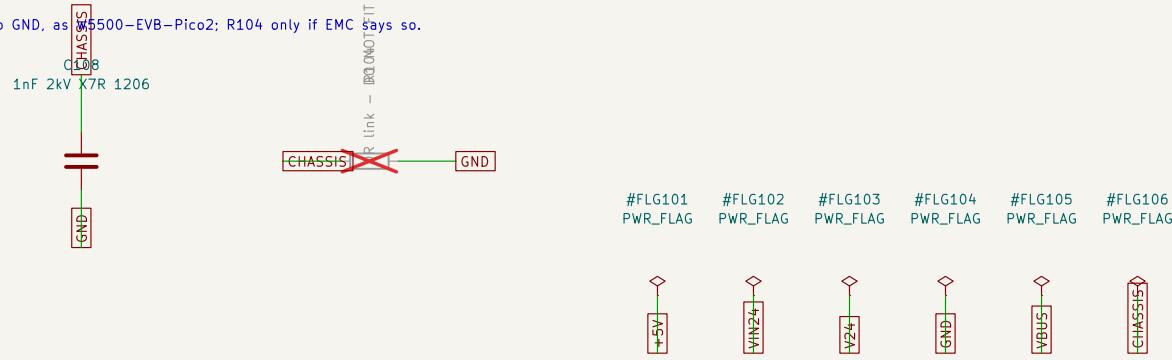


RAIL INDICATORS

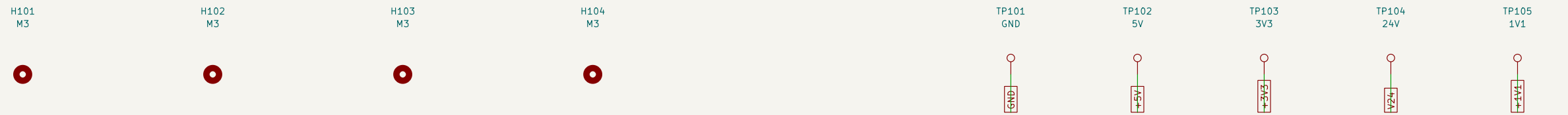


CHASSIS AND POWER FLAGS

Magjack and USB shields are CHASSIS; the mounting holes are bare. 1 nF 2 kV to GND, as on S500–EVB–Pico2; R104 only if EMC says so.

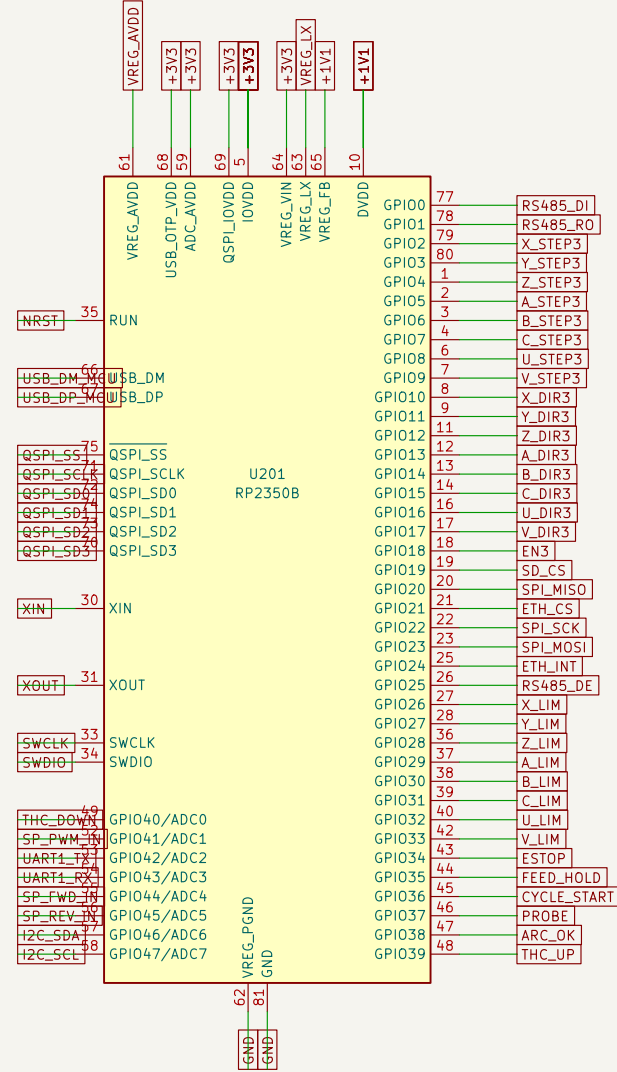


TEST POINTS AND MOUNTING



firmware/grblHAL/boards/my_machine_map.h. tools/check_pinmap.py compares them.

VREG_AVR00



Y201
12MHz ABM8-272-T3

XIN

XOUT_X

GND

15pF

15pF

1

2

3

4

5

6

7

8

9

10

11

12

13

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[illegible]

NRST also resets the W5500. 10k/1uF: ~10 ms, past its 500 us minimum.

The diagram shows two circuit sections. The first section for NRST includes a 3V3 supply, a 10k resistor (R203), the NRST pin, and a 1uF capacitor (C205) to ground. The second section for BOOTSEL includes a 3V3 supply, a 1k resistor (R204), the BOOTSEL pin, and a SW202 push-button switch to ground. A separate SW201 push-button switch is also shown connected to ground through a 1k resistor (R205).

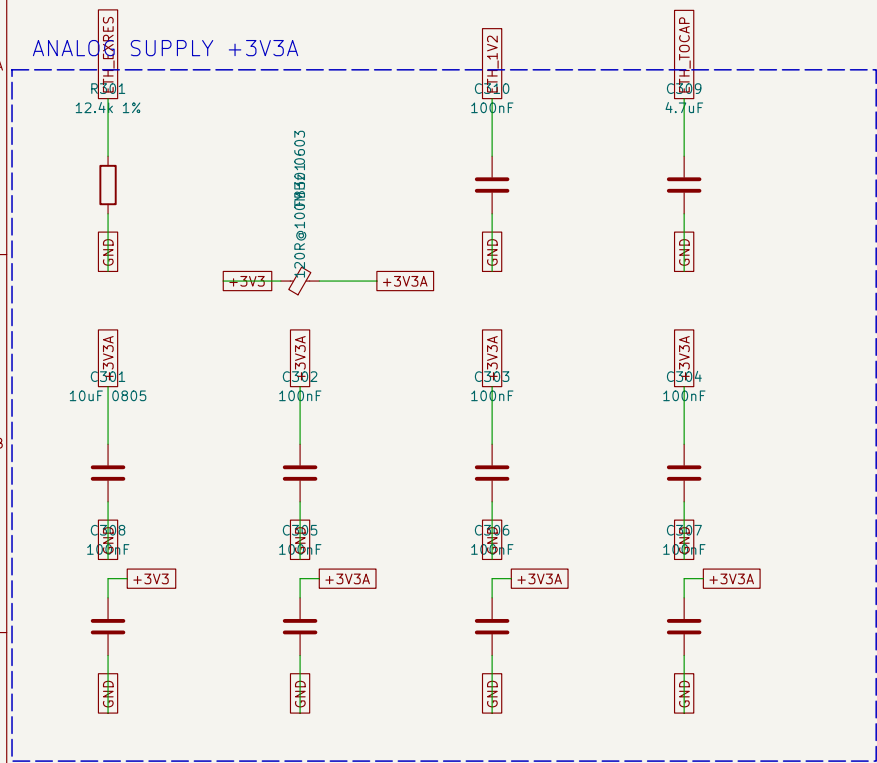
[illegible]

J202
SWD

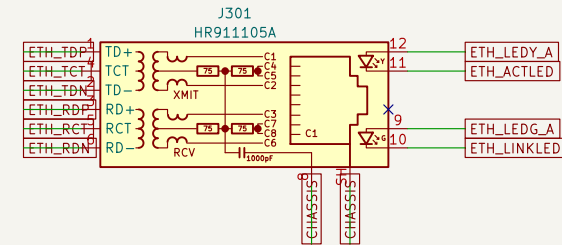
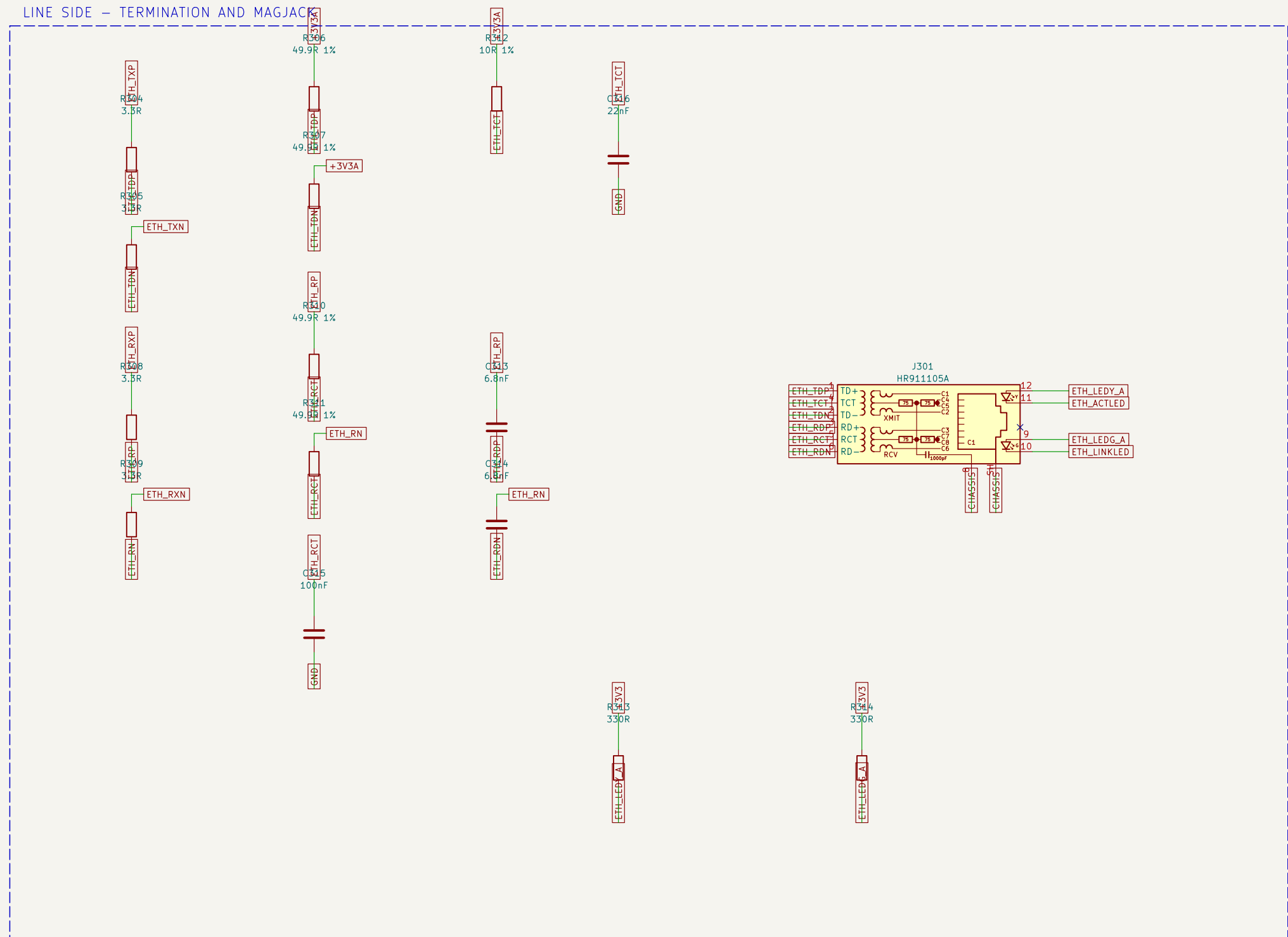
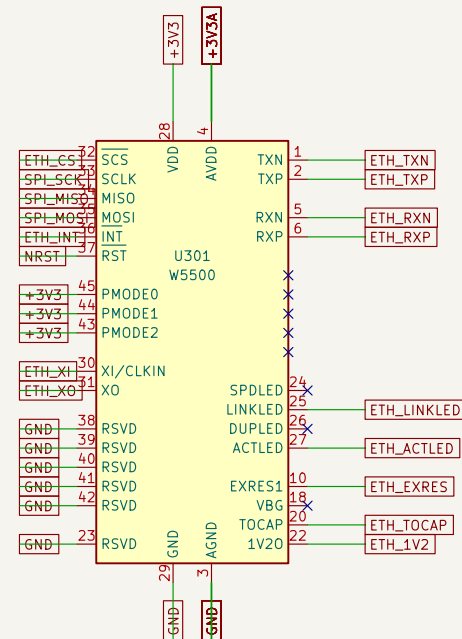
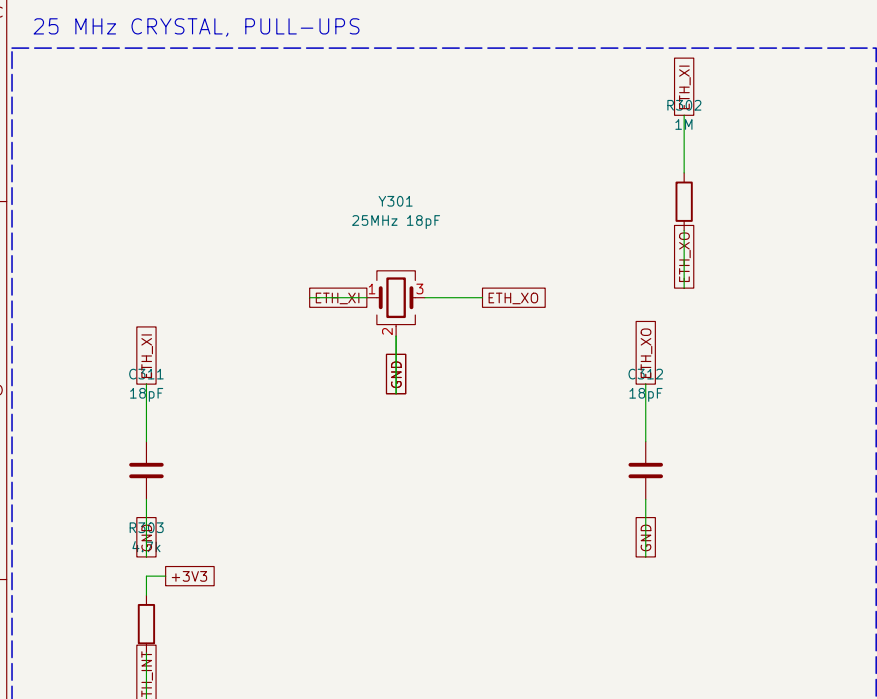
+3V3	1
SWCLK	2
SWDIO	3
GND	4

ETHERNET. W5500 on SPI0 (with the SD card). 10/100, Auto-MDIX.

Layout: 100 ohm differential pairs, short, no vias, nothing under the magjack.



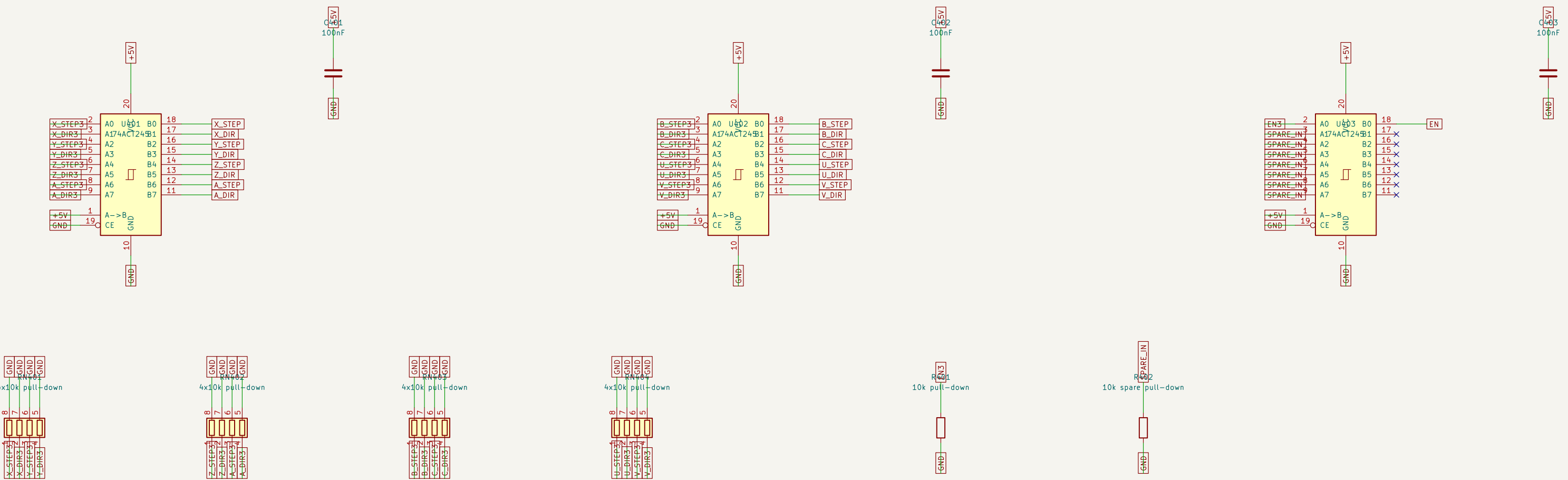
```
#FLG301
PWR_FLAG
```



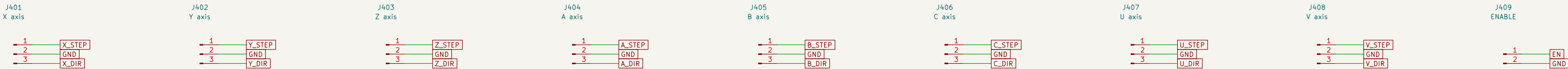
OUTPUTS. 3.3 V from the RP2350 -> 74ACT245 on 5 V -> terminals.

NOT ISOLATED, as A1: the axis terminals share ground with the board.

BUFFERS – 74ACT245, A->B, ALWAYS ENABLED



AXIS TERMINALS – PULSE / GND / DIR, AND ENABLE

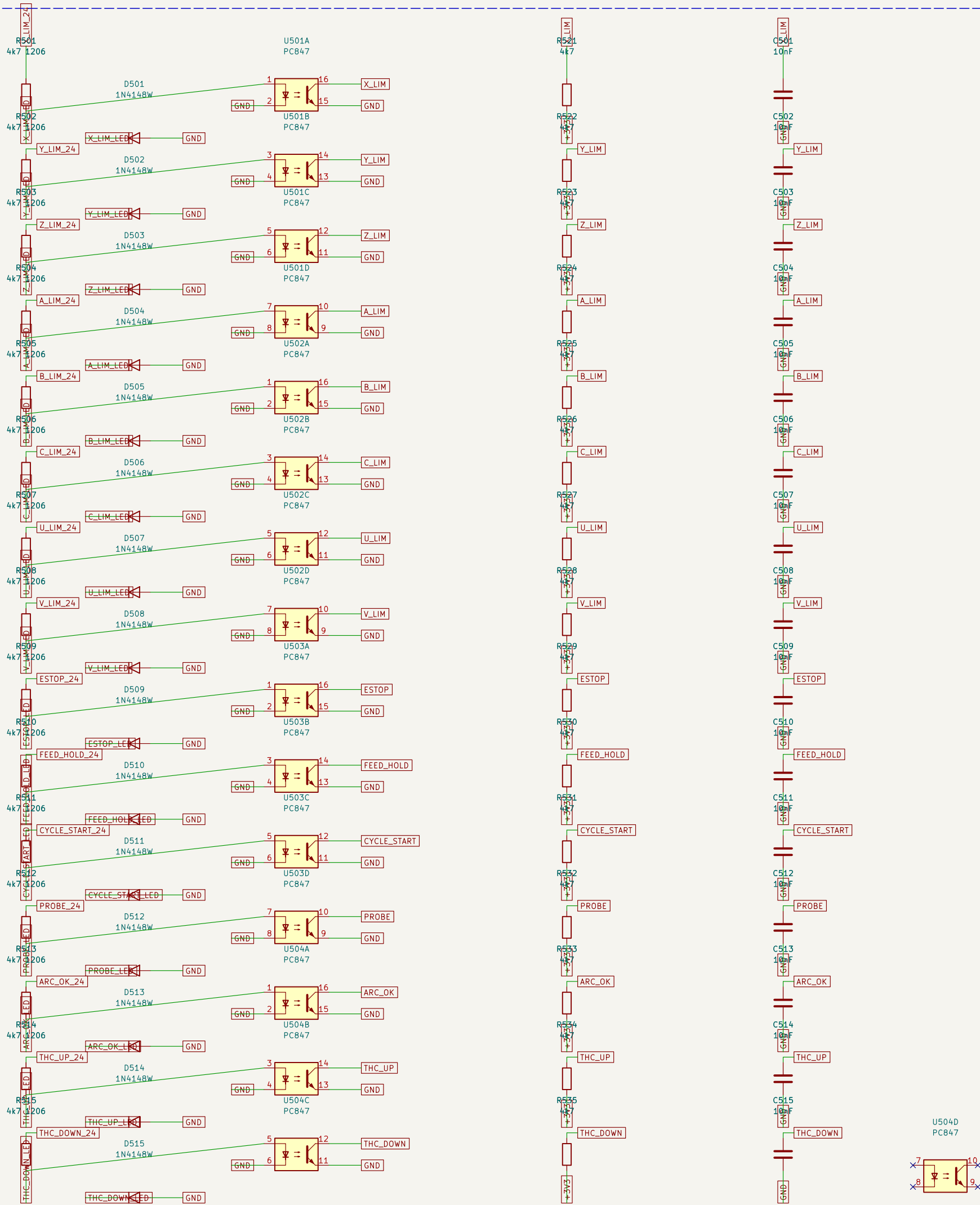


INPUTS. 24 V PNP sensors -> PC847 -> RP2350, active low.

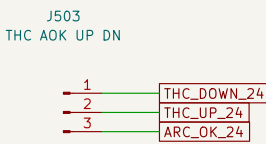
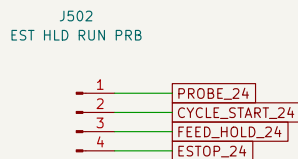
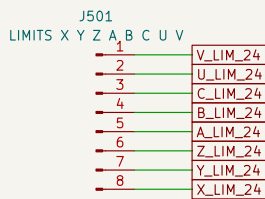
An anti-parallel diode across EVERY LED, as A1: the LED is rated 6 V reverse.

4k7 pull-ups to +3V3; strong enough for RP2350 erratum E9.

24 V INPUT CHANNELS



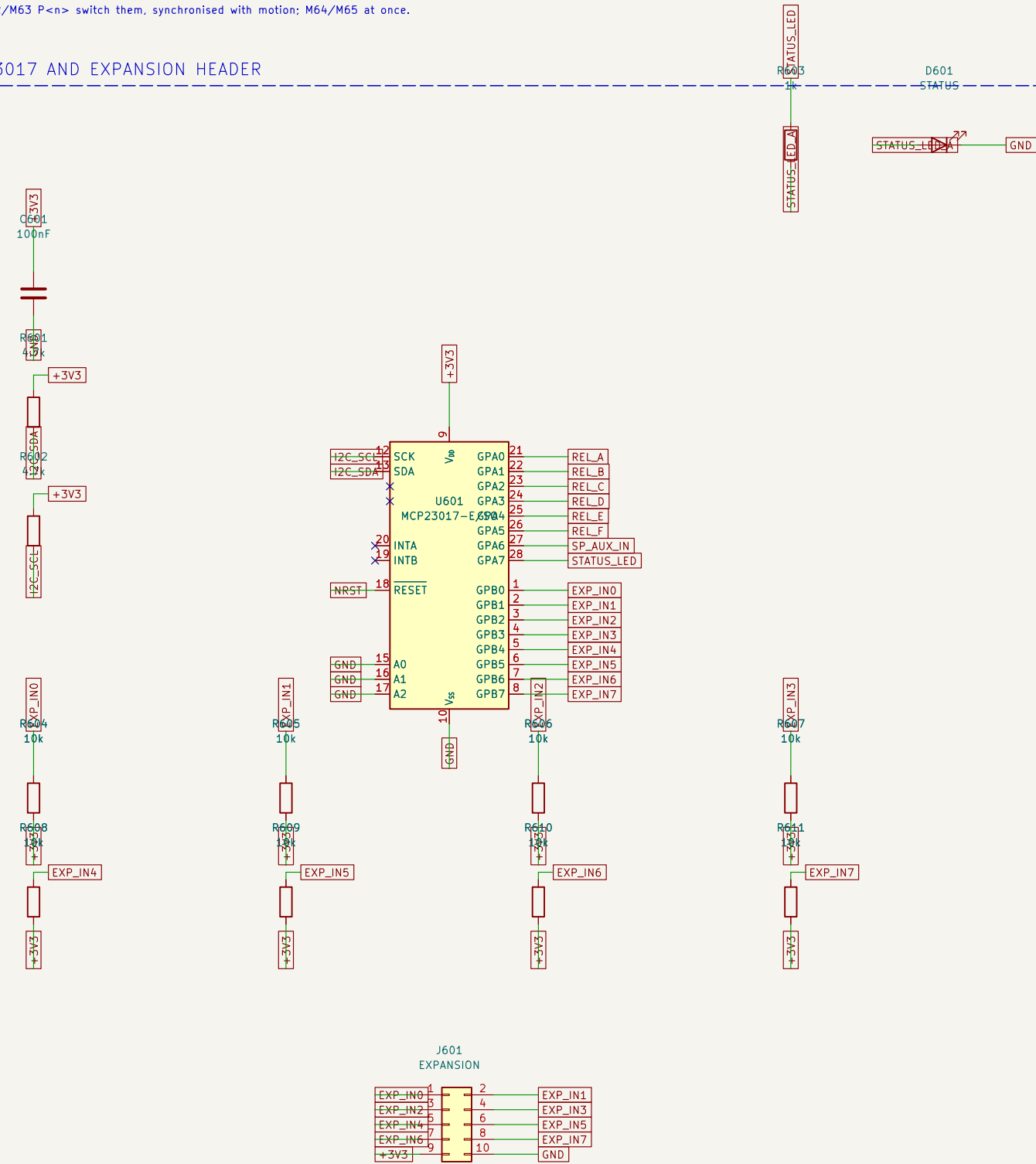
TERMINALS - SENSOR +24/0V FROM J101



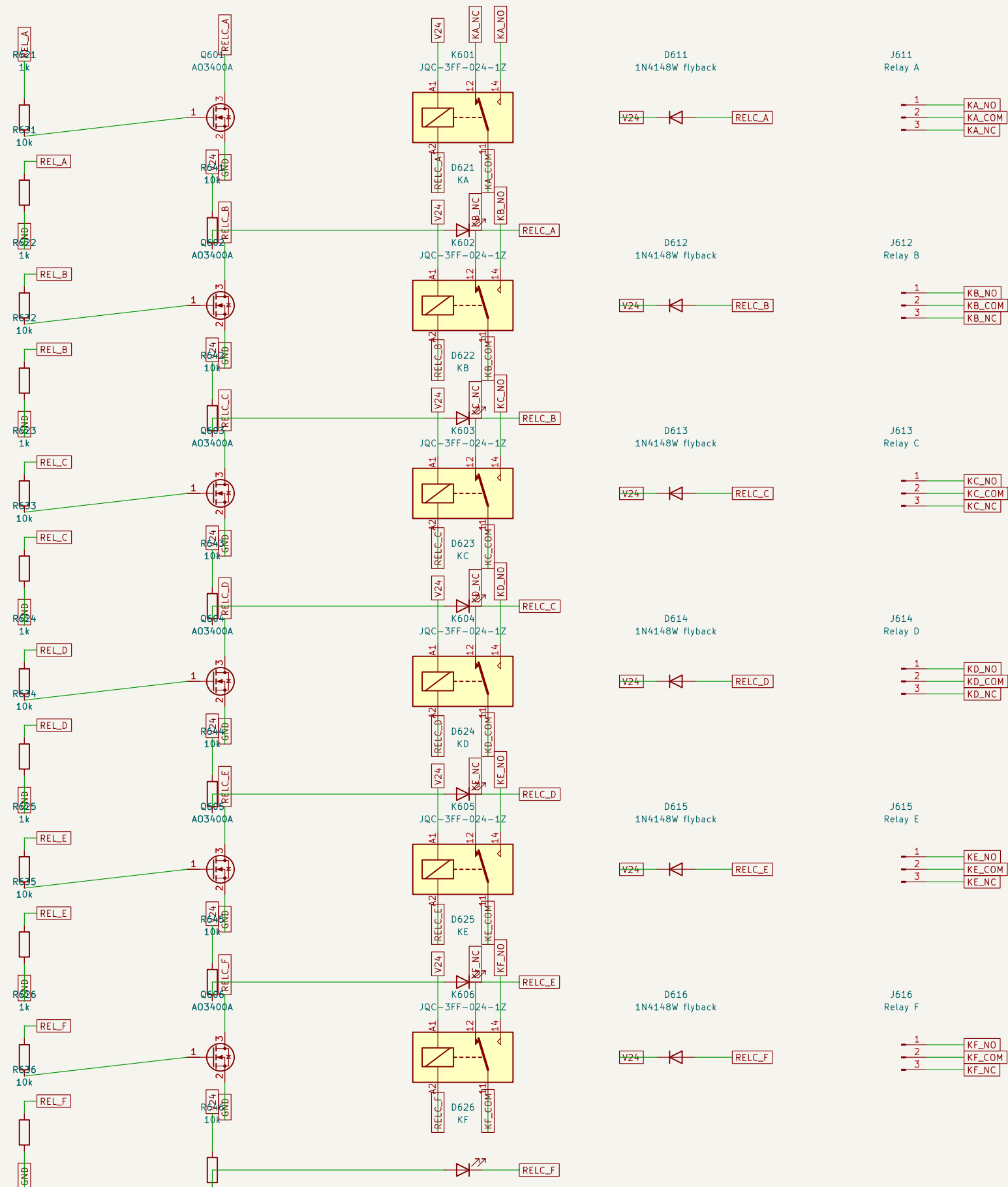
RELAYS. MCP23017 (I2C 0x20) -> A03400A -> 24 V relay coils.

G-code M62/M63 P<n> switch them, synchronised with motion; M64/M65 at once.

MCP23017 AND EXPANSION HEADER



SIX RELAYS – 24 V COILS ONLY, AS A1



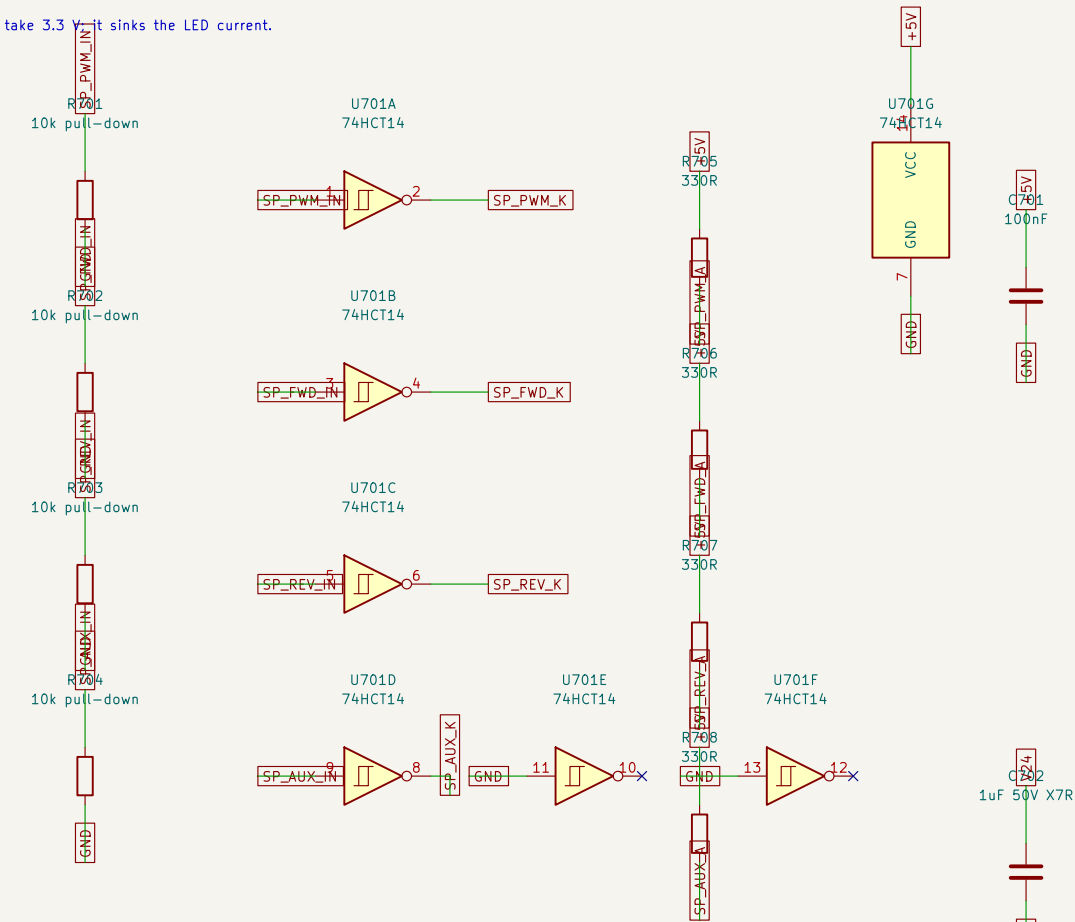
SPINDLE. 0-10 V and FWD / REV / AUX for the VFD, isolated from this board.

grblHAL: \$33 PWM frequency 100 Hz (as A1). Trim RV701 for 10.0 V at full speed.

BOARD SIDE – GPIO TO THE LEDS

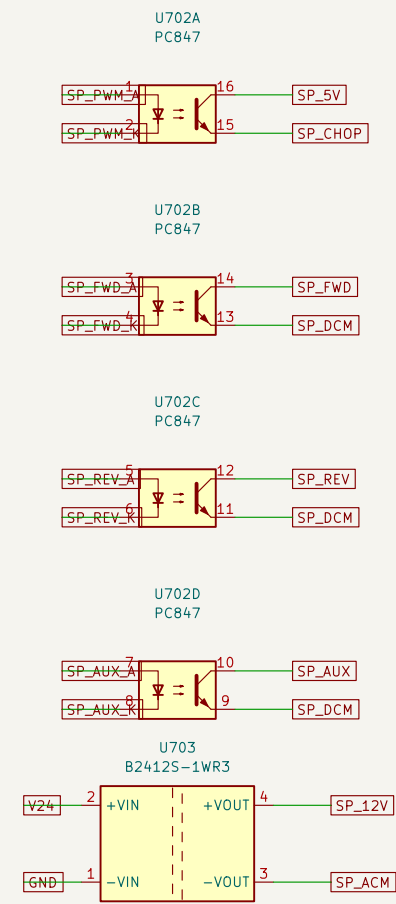
Pulled down, so with the RP2350 in reset every LED is dark.

74HCT14: TTL thresholds take 3.3 V; it sinks the LED current.

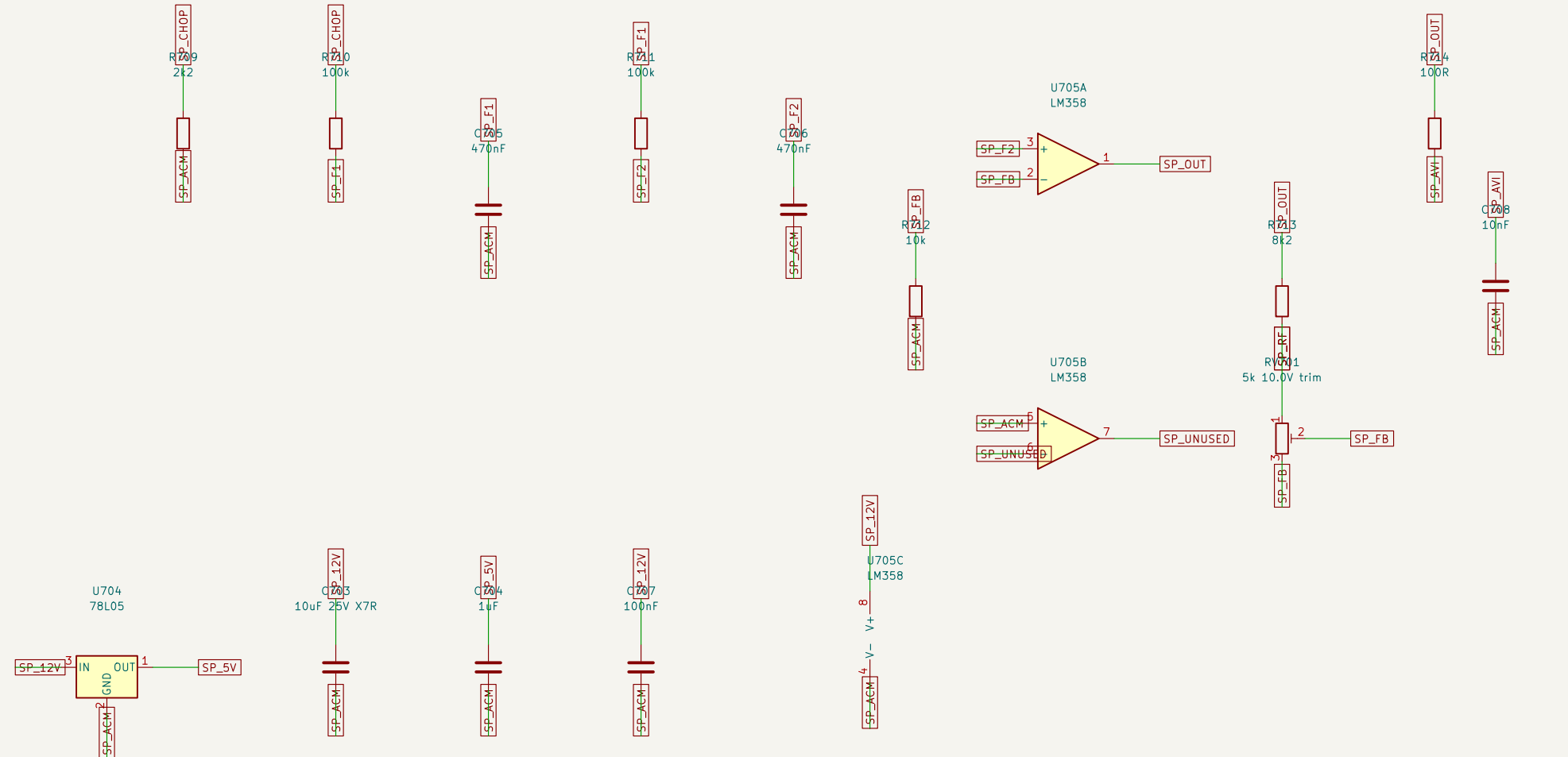


BARRIER

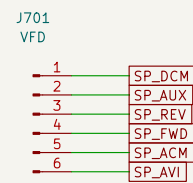
Nothing crosses this box but light and the DC-DC module.



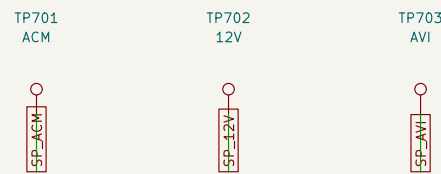
VFD SIDE – ISOLATED, ITS GROUND IS THE VFD ACM



VFD TERMINAL – AVI ACM FWD REV AUX DCM, AS A1

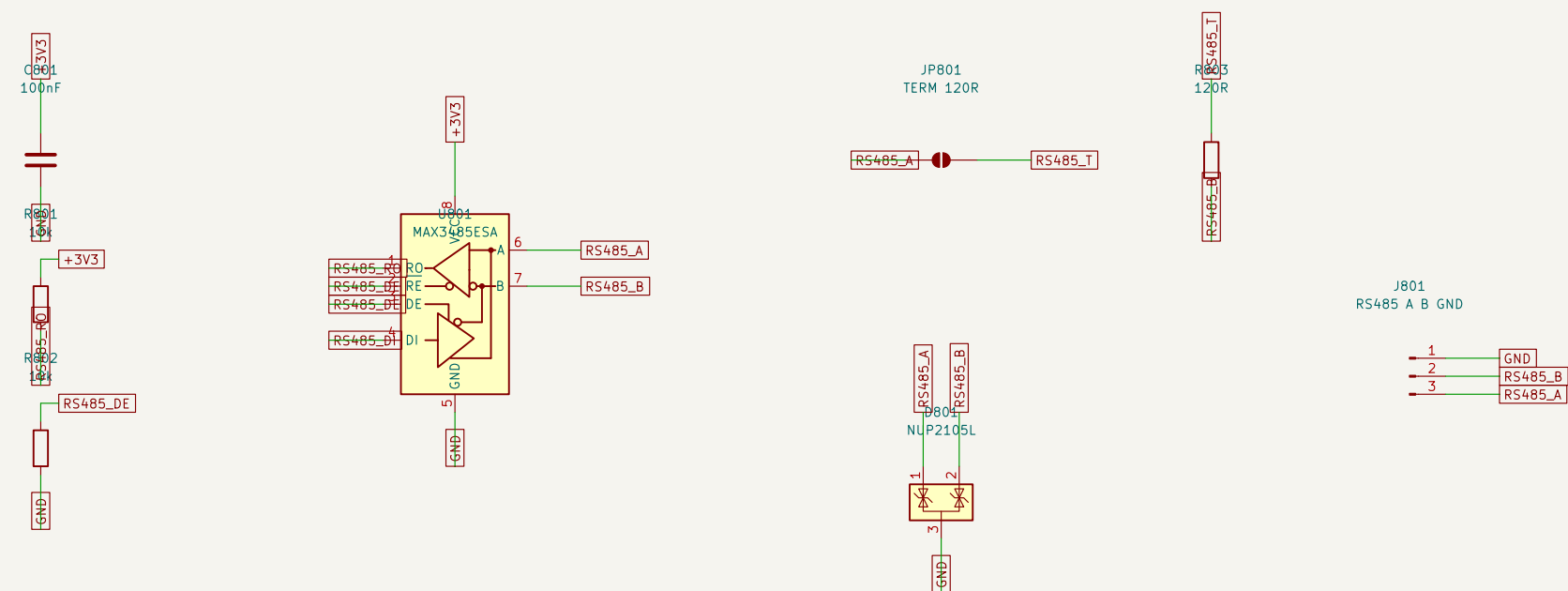


VFD-SIDE TEST POINTS – measure against ACM



RS485 VFD PORT AND HANDWHEEL

RS485 – MODBUS RTU TO THE VFD (UART0)



HANDWHEEL MCU – ATmega328P, 3.3 V, 8 MHz, UART1 TO THE RP2350

